Pivotum

THE DEGREE RISK INDEX · FALL 2026 EDITION

So you're thinking about engineering

The short version — written for you, not about you.

Everything that matters, none of the bits written for your parents.

We scored twenty-seven careers on how exposed they are to AI.

Licensed, site-based engineering: **4.0** out of 10.

Entry-level software development, for basically the same kind of mind and much the same maths: **8.1**.

That's the most decision-relevant comparison in the whole index if you're good at maths and like building things.

The 60-second version

- Engineering is one of the strongest technical bets we found.
- There are roughly three open engineering jobs for every qualified candidate.
- CS graduates currently face 6.1% unemployment. Engineering doesn't have that problem.
- **The discipline and the setting matter more than the school.** Site work scores 4.0; desk work 6.3.
- Electrical engineering may be the single most demand-secure choice in the index — and the reason is AI itself.

Engineering vs computer science

Worth seeing side by side, because almost nobody puts it this way.

	SOFTWARE	ENGINEERING
Is there a licence?	No	Yes — the PE stamp
Does it need you physically somewhere?	No	Yes — sites, commissioning
Who's liable if it fails badly?	The company	You, personally
Entry-level score	**8.1**	**4.0** licensed
Graduate market	6.1% unemployment	~3 jobs per candidate
Starting salary	~$87,000	~$81,000

The salary gap at entry is about $6,000 — smaller than most people assume. Software pulls ahead later. But it pulls ahead for the people who get in, and that's exactly the part that changed.

Software has two of our six protection factors sitting at zero for *everyone* in the profession, permanently. Engineering has neither of those gaps at the licensed end.

Why it's protected

The stamp is a legal monopoly. Certain designs must be signed by a licensed engineer, and personal liability follows that signature. No amount of AI capability transfers legal liability — that's a law thing, not a technology thing.

You have to physically be there. Site visits, commissioning, inspection. Reading actual conditions against what the drawings claim.

Every site is genuinely new. Not because the maths is hard — maths is what machines do well — but because a bridge sits on particular ground, in a particular climate, under a particular code, built by a particular contractor. That combination has no precedent even when every individual piece does.

The scores

DISCIPLINE	SCORE
Electrical power & grid	3.9
Civil / structural (PE, site)	4.0
Mechanical (industrial)	4.4
Manufacturing / process	4.9
Design & analysis (desk, CAD)	6.3

The line falls between the site and the office. Everything physical and accountable sits below 5. Everything at a desk climbs.

Which means the choice that matters isn't really *engineering or not*. It's *which kind*.

The electrical thing

This is the strangest finding in the profile and worth knowing about.

AI data centres are being built at record pace. Lawrence Berkeley National Laboratory estimates they could consume **325 to 580 terawatt-hours by 2028 — 6 to 12% of all US electricity.**

That has to be generated, transmitted and connected. By electrical engineers.

So electrical engineering is a career that's protected from AI and simultaneously being pulled by it. There isn't another one in the index like that.

What AI is actually taking

CAD drafting. Standard calculations. Simulation setup. Compliance checking. Generative design that produces fifty options in the time it used to take to draw one.

All of which used to be the junior engineer's job.

So the value moves upstream — toward specifying the problem and judging whether the answer applies here, and away from producing the drawing. Good news if you want to be responsible for outcomes. Less good if you picked engineering because you like drawing things.

The open question nobody's answered

Here's something the profession hasn't worked out, and we'd rather tell you than pretend.

Getting your PE licence requires about four years of supervised experience. That requirement is genuinely protective — firms *have* to employ and train juniors, which is why engineering's on-ramp is in better shape than software's.

But what you do during those years has changed. If you spend them directing generative tools and checking output, you log the same hours as someone who spent them running calculations by hand. Are you building the same judgment?

Nobody knows yet. The licence certifies that you did the time and assumes the time teaches.

Worth being deliberate about it: seek out the work that's genuinely hard rather than the work that's quick.

What we won't soften

The degree is hard and a lot of people leave. Engineering has among the higher switch-out rates of any major, and it's concentrated in the first two years of maths-heavy foundations. If maths is a struggle rather than a strength, that's worth being honest about now.

Licensure takes years. Four supervised years after graduating, plus exams.

It's often less creative than people expect. A lot of professional engineering is codes, standards, documentation and coordinating people who disagree. If you want creative latitude, architecture is closer to that.

Site roles can mean relocating, especially early and especially in civil.

Does this actually sound like you?

It suits people who:

- Are **comfortable being responsible when something physical fails** — the defining trait
- **Like constraints.** Engineering is solving problems inside hard limits — cost, material, code, physics — rather than open-ended creation.
- Are **precise and don't mind being checked.** Your work gets reviewed, stamped and audited.
- **Want the thing to exist afterwards**

Harder if you want creative freedom, dislike regulation and paperwork, or find sustained maths-heavy foundations a grind rather than a challenge.

What actually matters when picking a course

- **Accreditation for professional licensure.** Non-negotiable. Check it directly.
- **A placement year — and whether it includes actual site work**, not just office work
- **Whether they teach generative design tools *and* how they fail**
- **First- and second-year attrition rates**, and what they say about why people leave

And the discipline matters more than the school. Electrical and civil are in the strongest positions right now.

Two things worth doing

Visit a site. A construction site, a plant, a substation. An afternoon there will tell you more about whether this suits you than any brochure or open day.

Then find a working engineer and ask what the graduates on their team actually do all day, and how that's different from when they started.

And one more thing, if you're currently leaning toward computer science

This is worth an actual hour of thinking rather than a passing glance.

Same aptitude. Same maths. One has a legal monopoly, physical site work, personal liability and three open jobs per candidate. The other has none of the first three and 6.1% graduate unemployment.

That's not an argument against software. Plenty of people should do it, and the senior end is well protected. It's an argument that the comparison deserves more than the twenty seconds it usually gets.

If you want to go further — three things worth twenty minutes

- **Engineering hiring trends 2026** — where the three-jobs-per-candidate figure comes from, and which disciplines are hottest.
- **Engineering jobs, class of 2026** — salaries and growth by discipline, plus the fact that nearly half of working US engineers are 50 or older.
- **Is electrical engineering a good career in 2026** — the data-centre demand story in detail.

There's a longer version behind this — full scores by discipline, how they were worked out, the honest downsides, what to ask a program, and where we might be wrong. Your parents have it.